Design and Implementation of Carry Look-Ahead Adder using PTL with Comparative Analysis Against Transmission Gate Logic

Chezhiyan. M
Electronic Engineering (VLSI Design and Technology)
Chennai Institute of Technology Chennai
Tamil Nadu, India
chezhiyanm.vlsi2023@citchennai.net

R. M. Bommi
Department of Electronics Engineering
(VLSI Design and Technology),
Chennai Institute of Technology Chennai,
Tamil Nadu, India
bommirm@citchennai.net

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***Abstract*— In modern VLSI design, speed, power consumption, and area are key constraints of modern-day VLSI design. There exist several styles of logic that address these design constraints, including Dynamic CMOS, Static CMOS, Reversible Gates, Transmission Gates (TG), Pass Transistor Logic (PTL), and new- fangled ones like Gate Diffusion Input (GDI). The Implementation and design of a 4-bit Carry Look-Ahead Adder (CLA) employing Pass Transistor Logic (PTL) and the Transmission Gate (TG) style are presented here. PTL is effective in reducing the number of transistors, thus producing a smaller area and lower power consumption, hence it is used for low-power and small- area design. PTL reduces the number of transistors by 58.82% and optimizes the power consumption by 28.14% compared to TG. However, TG is better in terms of performance speed and delay time, reporting a 48.5% improvement over PTL in delay, providing greater signal integrity and more speed and reliability. Cadence Virtuoso was used to simulate and model all the designs to accurately analyze transistor count, delay, and power consumption.**

Keywords*— PTL-Pass Transistor Logic, TG-Transmission Gates, GDI – Gate Diffusion Input, CLA-Carry Look Ahead Adders.

I. INTRODUCTION

In the field of Very Large Scale Integration (VLSI), adders are the basic fundamental components used in a wide range of applications, which includes Arithmetic Logic Unit (ALUs), Digital filtering, Signal processing and Processor cores[1]. The efficient of an adder directly impacts the overall performance, area and power consumption of the VLSI systems and its production [2][3] . The Carry Lookahead Adder (CLA) is a vital design in high-performance VLSI systems because of its ability to overcome the delay in the traditional adders like Ripple Carry Adder[4]. It is achieved by computing the propagation delay significantly. Speed is an important factor for Carry Lookahead Adders (CLAs) due to their widespread use in high-performance VLSI applications. Systems in modern days require higher clock frequencies, additional operations to be executed in fewer clock cycles. In real-time systems like video processing and digital signal processing (DSP), immediate results are essential to maintain functionality, whereas in latency sensitive applications like networking and autonomous vehicles, delay can directly impact system responsiveness. Additionally, reducing the area of the CLAs is essential to optimize silicon utilization, reduce parasitic capacitances and lower fabrication costs, which indirectly increases the speed and power efficiency [5]. Using Pass Transistor Logic (PTL) instead of conventional CMOS logic is an effective way to achieve the minimum number of transistors in a circuit [6][7] Whereas using Transmission gates instead of PTL provides full swing voltage at the output, which overcomes one of the main limitations of the PTL [6].

II. CARRY LOOK-AHEAD ADDER DESIGN

Carry Look-Ahead Adder or CLA is a method for digital electronics that implements a binary addition in the least amount of time. Instead of the speed-restricted time taken by a Ripple Carry Adder (RCA), it reduces the amount of time the carry bits take to travel across the stages [4][5]. It is based on Generate (G) and Propagate (P) signals for its operation called 'Parallel Carry Computing' [8]. This eliminates the sequential delays present with traditional adders as carry values are precomputed simultaneously for all stages [5].

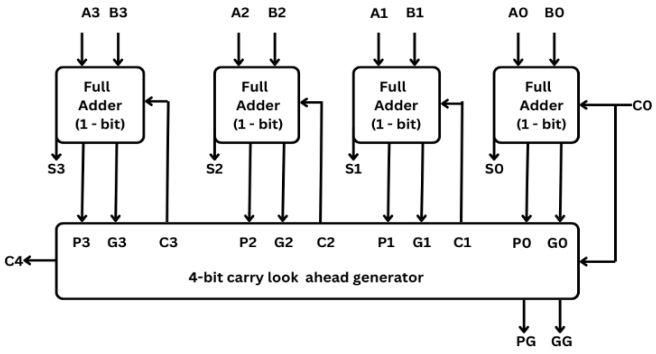


Fig. 1. 4-bit CLA Block diagram

Carry Generator Circuit The carry generator circuit is a very important constituent of a CLA. It assists in determining whether to generate carry at a bit position by using generate and propagate signals. This way, Generate signals help create a carry directly from the input bits and propagate allows an incoming carry to be passed through. The circuit simultaneously pre-computes all the carry values for all bits. This means that there is no sequential propagation in the addition process; therefore, it makes the operation faster.

Gi - Generate Signal:

$$\mathbf{Gi = Ai.\ Bi} \qquad (1)$$

Case i): If Gi = 1, A carry is generated if both Ai & Bi are 1.

Case ii): If Gi = 0, From this bits no carry is generated.

Pi – Propagate Signal:

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$$Pi = Ai \oplus Bi \qquad (2)$$

Case i): If Gi = 1, A carry is generated if either Ai & Bi are 1.

Case ii): If Gi = 0, No carry is propagated.

Sum Output :

$$Si = Pi \oplus Ci \qquad (3)$$

Carry Output :

$$Ci+1 = Gi + Pi.Ci \qquad (4)$$

By applying this equation (4), CLA eliminates the dependency of previous stages' calculations. This capability of parallel computation makes arithmetic operations much faster and more efficient [8]. Therefore, the CLA is used in processors and high-speed computing systems for improving performance.4-bit CLA Boolean function equation [5].

$C1 = G0 + P0.\ Cin$

$C2 = G1 + P1.C1$

$= G1 + P1.\ G0 + P1.P0.Cin$

$C3 = G2 + P2.C2$

$= G2 + P2.\ G1 + P2.P1.G0 + P2.P1.P0.Cin$

$C4 = G3 + P3.C3$

$= G3+P3.G2+ P3.P2.G1 + P3.P2.P1.G0+P3.P2.P1.P0.Cin$

III. Carry Look-Ahead Adder Gate Level Design and Truth Table

The CLA circuit is designed with three important gates they are XOR, AND, OR logic gates. The above circuit diagram shows the Parallel Carry Computation [8] of a 4-bit Carry Look-Ahead Adder. The CLA circuit eliminates ripple carry delay [4] thus gives rapid carry computation. Thus, it is appropriate for high-speed arithmetic operation. It produces carry propagate (P) and carry generate (G) signals with the help of XOR and AND gates respectively. These are then combined together, through OR gates, to compute carries for all bits in parallel [9]. The CLA adder accomplishes better performance by lowering the propagation delay, enabling quicker addition operations, and is commonly utilized in arithmetic units of microprocessors and digital systems that need quick computation.

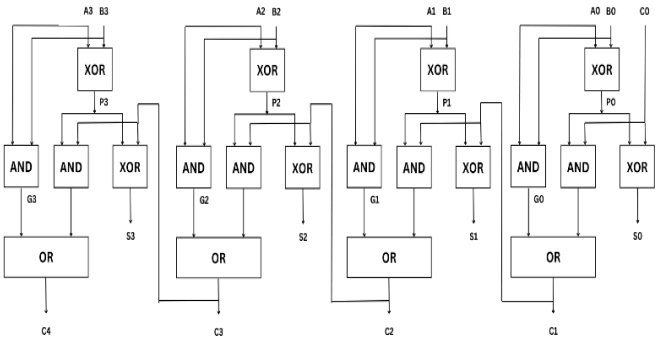


Fig. 2. Parallel Carry Computation for 4-bit CLA

TABLE I. TRUTH TABLE FOR FULL ADDER

A	B	C	SUM	CARRY
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

TABLE II. TRUTH TABLE FOR A 4-BIT CLA

A	B	C	C i+1	STATUS
0	0	0	0	No Carry Generate
0	0	1	0	
0	1	0	0	
0	1	1	1	No Carry Propagate
1	0	0	0	
1	0	1	1	
1	1	0	1	Carry Generate
1	1	1	1	

This table shows how the Carry Look-Ahead Adder correctly computes carry bits in an efficient manner that enhances the speed performance compared to the simple binary adding method. The inputs to the system are A, B and C. In the system, A and B are the summands and C, the carry to be added as input from the previous stages [5]. C i+1 represents the carry out of the output, and "STATUS" indicates when a carry is generated or passed. For the truth table, there are three possible cases covered. In the first case, regardless of C if A and B are equal to 0, nothing is carried out. In second case, if A is not equal to B, and C is equal to 1, carry the result from the previous step. This CLA always has a carry in A and B, with the value 1 in it, and this design significantly accelerates the speed of addition compared with the ripple-carry style addition by reducing the delay incurred when propagating carries. So, this CLA is an inevitable component in high-speed digital circuits [5].

IV. Pass Transistor Logic (PTL)

Pass transistor gate level design is considered as one of the digital circuit designs where it reduces the number of transistors by using them to pass logic levels directly between nodes. Unlike conventional CMOS logic, where pass transistors are switches controlled by the gate signals to allow or block input signals from going to the output, the method has advantages of small area, low power consumption, and fast operation since the capacitance is reduced [7] . However, some problems are voltage degradation with threshold voltage drops and a limited fan-out which, in most cases, need restoration circuits. Pass transistor logic is applied in most cases in multiplexers, arithmetic circuits and transmission gates where it enables them to be very compact [6].

A. PTL Full Adder implementation in cadence

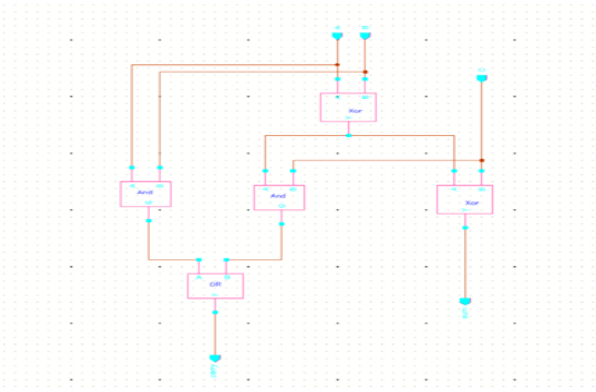


Fig. 3. Full Adder circuit design in Cadence

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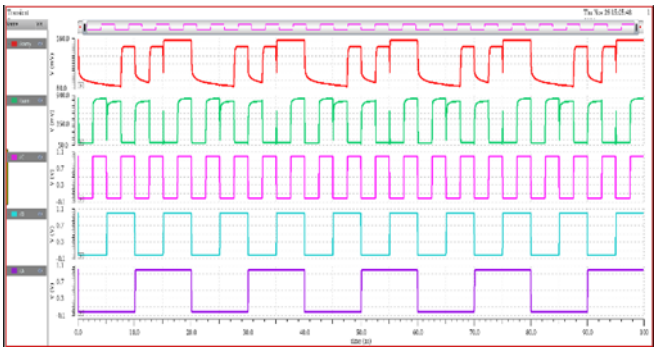


Fig. 4. Output Graph of Full Adder circuit (Sum, Carry)

The PTL-based full adder waveform (Fig. 4) is also seen to be correct with spikes and falls due to voltage Vth drop, charge sharing, and parasitic capacitance, which exhibit poor fan-out and lead to signal integrity loss. Further information regarding parasitic capacitance was presented in these papers [10][11][12]. The problems occur as a result of the decreased driving strength and feeble output stages in PTL circuits, particularly when cascading several stages. The addition of buffers, parasitic reduction, or hybrid styles [6] can offer stable signals but retain the low power and small-sized benefits with PTLs. Adding Transmission Gate (TG) logic in key nodes also increases signal stability and performance without an area penalty for PTL, thus providing an optimal mix for high-speed, low-power arithmetic circuits.

B. PTL Full Adder implementation in cadence

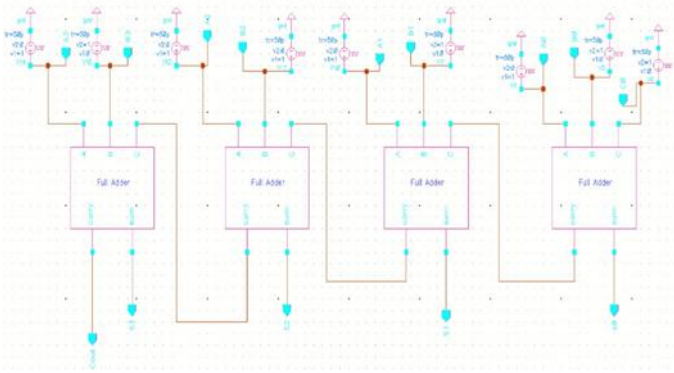


Fig. 5. CLA circuit design (PTL) in Cadence

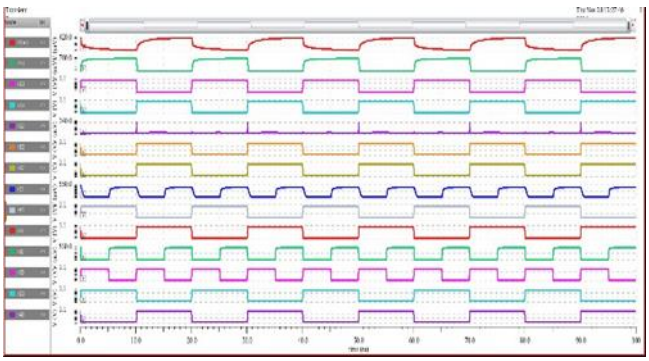


Fig. 6. CLA output waveform in Cadence

This is a time-domain simulation (Fig 6) of the Carry Look- Ahead Adder based on the PTL. The waveform displays the output and input of the circuit with respect to time. The lower traces are the binary inputs A, B, and carry-in Cin, and the middle traces are the sum outputs S0, S1, S2, S3, and the top trace is the carry-out Cout. Some deterioration in voltage level at outputs is observed due to this inherent fall in threshold voltage Vth of the PTL [6], which states that an nMOS transistor sourcing a logic '1' causes an event peak below VDD volts. Also, some marginal slowing down of transitions mirrors on propagation delays due to comparatively poor drive strengths in PTL implementations. Flattening of noise and signal in certain traces indicate a loss of voltage swings and crosstalk effects that may affect signal integrity [6]. Although the CLA design functions, degradation along with timing suggests compromises in PTL and requires restorative circuits to function in reliability. Additionally, increasing transistor sizing or inserting buffers would reduce the weak drive to some extent, but at the cost of increased area and power cost. Future enhancement could also seek hybrid approaches utilizing TG or GDI logic in the critical paths to achieve the best performance and signal stability. Hybrid implementations of this nature may not only address voltage degradation but also improve energy efficiency. Full transistor-level optimization considering layout may be able to further reduce parasitic effects, improving the transmission of signals and reducing noise-related issues. Also, advanced modeling techniques may be used to predict signal integrity problems more correctly at the design phase.

V. TRANSMISSION GATES

The Transmission gates (TG) are also known as analog switches. It is an essential component in digital design and specifically an implementation for switches. It contains two complementary field-effect transistors [13][14]: NMOS, a negative-channel MOSFET; PMOS, a positive-channel MOSFET. Unlike discrete MOSFETs, the substrate, or bulk, of a transmission gate is not internally connected to its source. Instead, the NMOS and PMOS are connected in parallel, while the common drain and source terminals are shared [13]. Their dual transistor structure allows for bidirectional signal flow with minimal voltage drop, making TGs highly efficient compared to single-transistor configurations, such as those in PTL. The other significant design differentiation of transmission gates is how substrate terminals are dealt with. The substrate terminals here again become hard tied to fixed supply voltages that make the parasitic substrate diodes have nothing to interfere with the working of them. For PMOS transistors, the substrate is connected to VDD, and for NMOS, the substrate becomes shorted to ground that remains negative and positive supply volts. Parasitic diodes are back- biased to keep unwanted current paths out and signal flows intact.

The unique structure of TGs provides better efficiency than PTL in some applications [7]. PTL uses NMOS or PMOS devices to implement logic functions, and it is often plagued by threshold voltage (Vth) losses, especially in cascading stages. Transmission gates, by combining complementary transistors, eliminate these losses and provide full voltage swing at the output, which results in better signal integrity and reduced power dissipation. In addition, TGs have a symmetric resistance characteristic, which is very useful for high-speed, low-power, digital circuits. Because of efficient operation, TGs are the best solution for applications where minimal distortion of the signal, smaller propagation delay, and reduced power consumption are critical requirements. A comparison of the PTL and TG logic shows that it is often the case that TGs outperform PTL in applications requiring strong and reliable switching mechanisms which makes them invaluable in current VLSI design.

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The transmission gates, employing both complementary p-channel MOSFET and n-channel MOSFET, provide a virtually full rail-to-rail voltage swing in them [15]. The complementary transistors guarantee low-resistance paths for both the high, logic 1, and low, logic 0, voltage levels after the control signal permits the TG. This, thus, causes negligible voltage degradation through the gate and hence enables signal propagation without threshold voltage (Vth) loss. Carry look-ahead adders with transmission gates provide full swing, low delay, and high reliability; pass transistor logic, however, conserves area in terms of voltage degradation [7]. Hybrid design [6] involving both techniques (TG for paths sensitive to speed and PTL for area saving) offers all three aspects' optimization—speed, power, and area. Hybrid techniques have special applications where low power high performance is an issue and also area efficiency as system reliability critically depends on them.

A. TG Full Adder implementation in cadence

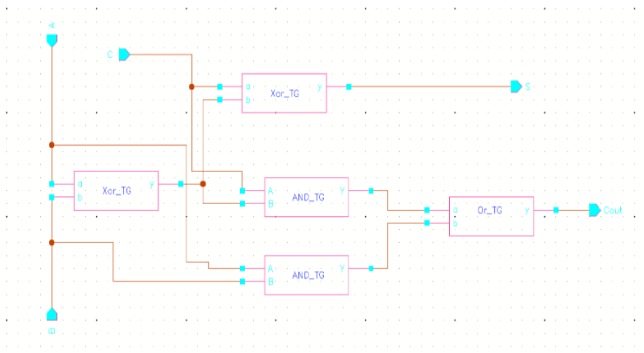


Fig. 7. Full Adder circuit design (TG) in cadence

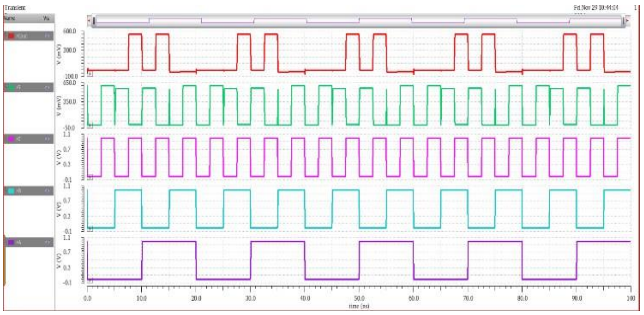


Fig. 8. Output Graph of Full Adder circuit (Sum, Carry)

B. TG CLA implementation in cadence

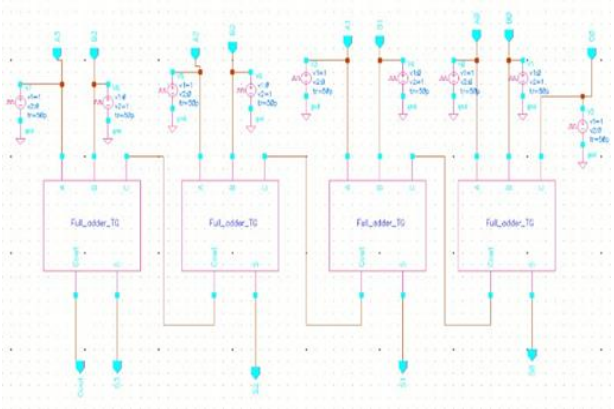


Fig. 9. CLA circuit design (TG) in cadence

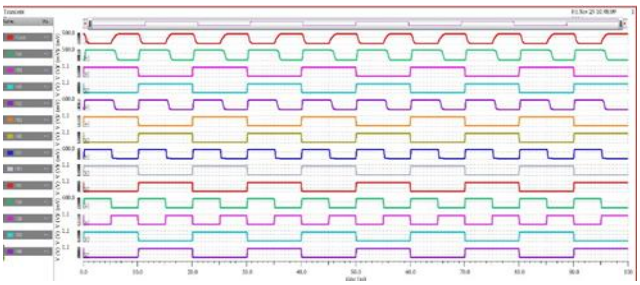


Fig. 10. Output Graph of CLA circuit (Sum, Carry)

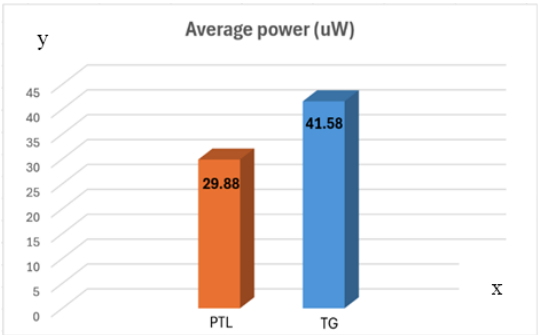
The carry look-ahead adder implemented with Transmission gates (TG) (Fig 11) provides full voltage swing, better signal integrity, and lower propagation delay. Thus, it is more efficient for high-speed and low-power designs compared to Pass transistor logic (PTL) [7]. PTL is area-efficient but suffers from voltage degradation due to threshold voltage loss, particularly in cascaded stages, which affects signal reliability. Bidirectional signal flow and low distortion because of complementary NMOS and PMOS structure characterize the TG-based adders [13][14]. The compact design of PTL can be beneficial in area-critical applications. Hybrid approach: using TG for critical paths and PTL for non-critical ones optimizes speed, power, and area.

VI. Result Analysis

The two implementations of the Carry Look-Ahead Adder (CLA) using pure Pass transistor logic (PTL) and pure Transmission gates (TG) show considerable differences in design efficiency. The metrics used were transistor count, time delay, and average power. Although PTL has fewer transistors, it has higher time delay and power dissipation due to threshold voltage losses and degraded signal integrity [7]. In contrast, with TG-based CLA there is no delay but will increase transistor count. Using a table and bar graph, analysis shows the case for high-performance applications. The Table shows that the PTL-based CLA implementation consumes only 56 transistors, much fewer than those required in the TG-based design, at 136. This reduces the number of transistors consumed in PTL and, thus, is more area-efficient. Thus, it can support more compact designs for applications that require fewer resources. Nevertheless, this advantage will result in signal degradation and increased delay, making TG more desirable for performance-sensitive circuits.

TABLE III. RESULT TABLE

Parameters	PTL	TG	Static CMOS [16]
Transistor Count	56	136	512
Average Power (μW)	29.88	41.58	51.33
Time Delay (ns)	2.35	1.20	1.13



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Fig. 11. : Transistor Count

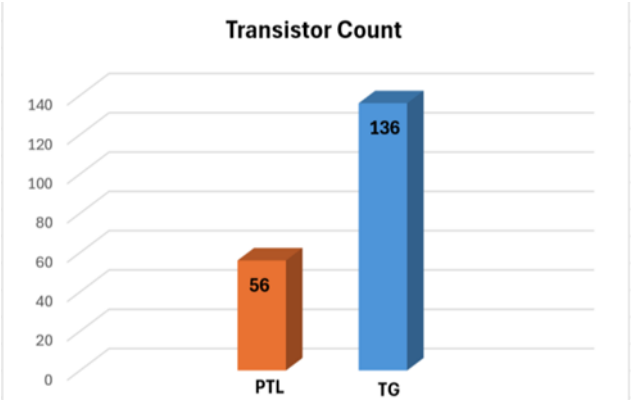


Fig. 12. Average Power Consumption

The average power consumption of the Carry Look-Ahead Adder, CLA, is much lower in the Pass Transistor Logic (PTL), at 29.88µW because of less transistor count and simpler circuit. The Transmission Gate (TG) takes 41.58 µW as average power dissipation on account of its higher number of transistors as well as full voltage swing, thus ensuring better signal integrity [8]. The time delay for Carry Look-Ahead Adder in PTL is measured as 2.35 ns that is significantly higher due to threshold voltage losses and signal propagation degradation. Conversely, the implementation of TG has achieved a delay as low as 1.20 ns, which leads to faster operation with less signal degradation and is most suited for high-speed applications.

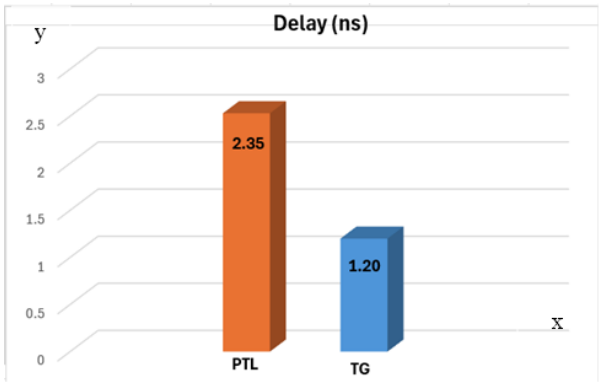


Fig. 13. Delay

VII. CONCLUSION

The analysis of the design and synthesis of a 4-bit CLA adder with Pass Transistor Logic (PTL) and Transmission Gate (TG) logic describes the pros and cons of both methods in VLSI design. PTL is a good methodological approach in area-efficient design since it consumes a smaller number of transistors, hence saving chip area and switching capacitance. It indicates less power dissipation and high-frequency operation in small-area designs. The reduction of threshold voltage, nonetheless, degrades the signal of the PTL, thus causing increased delay time and limiting its use in high-performance circuits. TG-based design, conversely, is encouraged by full-swing voltage, reduced propagation delay, and improved signal integrity and thus, finds itself more suitable for fast and dependable applications. However, this layout is growing transistor count, which results in increasing chip area and more power consumption in switching, thus it's not so optimal in resource-restricted designs.

Trade-off between speed, power, and area in CLA implementation is therefore required under a hybrid model that applies PTL to area-critical paths and TG to delay-critical paths. All the models have areas of application in VLSI design: PTL is best suited to small systems with low power requirements, while TG is best suited in high-performance systems. This versatility makes them remain relevant in VLSI applications.

By comparing, it is discovered that PTL and TG have different efficiencies in the three main parameters. PTL has better efficiency in reducing transistor number, with an efficiency of 58.82% as opposed to TG. It is hence ideal for designs that are compact. Yet, TG is superior to PTL in power and speed. PTL is also extremely power-efficient when it comes to average power consumption, while TG lags far behind with an efficiency ratio of only 28.14%, positioning PTL as the less power-consumption one. Moreover, TG also demonstrates unbelievable energy efficiency in delays, with a staggering 48.5% efficiency compared to PTL, to facilitate faster execution. PTL stands tall in compactness, while TG stands tall in energy efficiency and speed.

While this work discovers the merits and demerits of PTL and TG logic, potential future research might entail exploring a hybrid CLA architecture that unites Transmission Gate (TG) and Gate Diffusion Input (GDI) logic. GDI is a developing technique that has been acknowledged to reduce transistor count and power dissipation with assurance of performance. The union of TG and GDI might enhance speed and power efficiency for CLA architectures. The TG logic can be applied in critical timing paths for high-speed performance, while GDI logic can be applied to non-critical and area-constrained paths to conserve transistor count and power. More research and simulation of such hybrid designs may lead to an optimized CLA implementation with improved speed, power, and area balance, and thus making it more appropriate in low-power VLSI systems of next- generation.

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